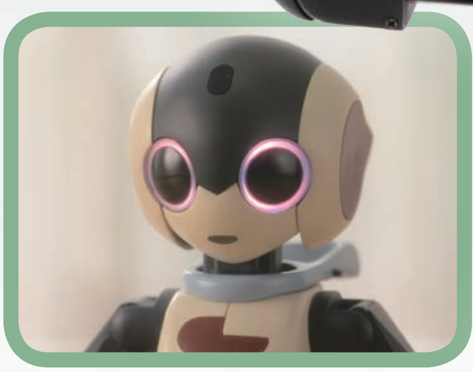
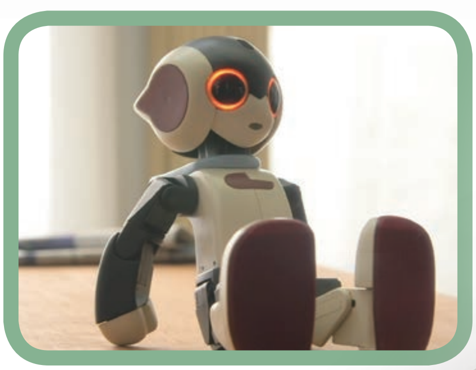
Build your own Robi

Pack 03





Stages 7-10: See Robi's head move for the first time

DeAGOSTINI MODEL SPACE

Build your own Robi

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DeAGOSTINI MODEL SPACE™

www.model-space.com

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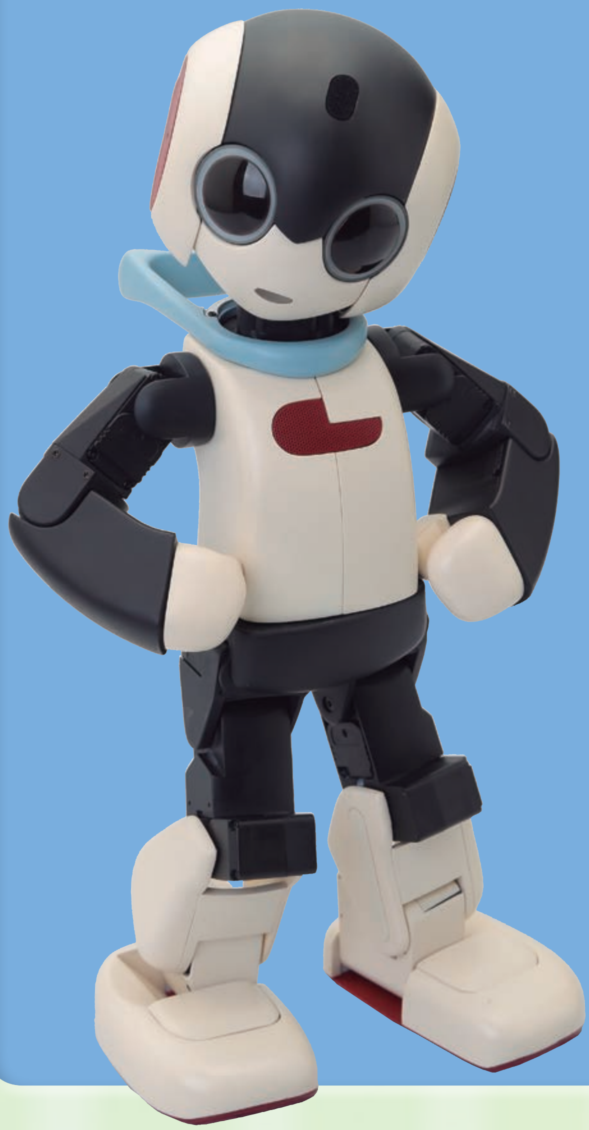
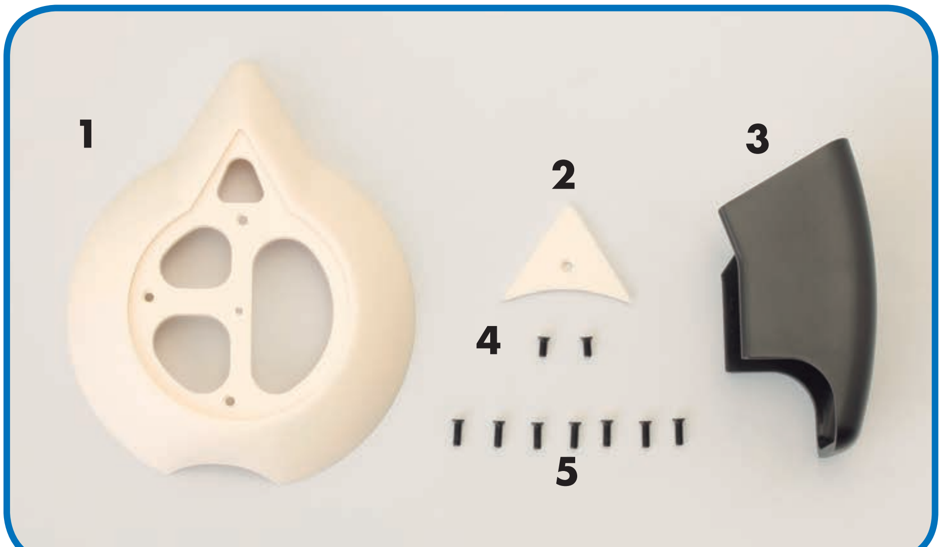


STAGE 7: CONTINUE ROBI'S HEAD, THEN TEST ITS MOVEMENT

In this stage you will take another step towards completing Robi's head by adding his left ear, together with the right ear you assembled in Stage 4. You will then test Robi's head movement using the neck stand and servo.

Once the ear is fitted to the right side of Robi's head frame, you will be at a point where you are able to see him turning his head. You will do this by fitting the head onto the shaft of the servo you connected in the previous stage, then running the same servo test as before. Make sure you align the parts correctly, and handle the assembly with care.

PARTS TO BE ASSEMBLED



YOUR PARTS

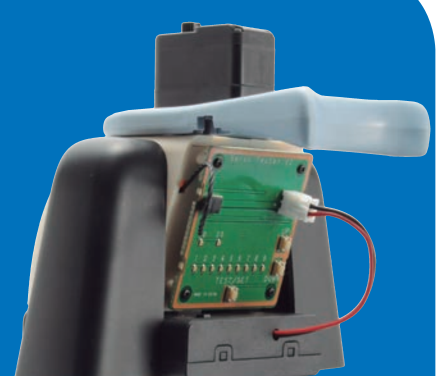
1 Left ear base
2 Left ear backing
3 Right forearm
4 M2 x 4.5mm countersunk screws (x 2)
5 M2 x 6mm countersunk screws (x 7)

SAVED PARTS

You will be using previously assembled parts for this session, so have these to hand before starting.

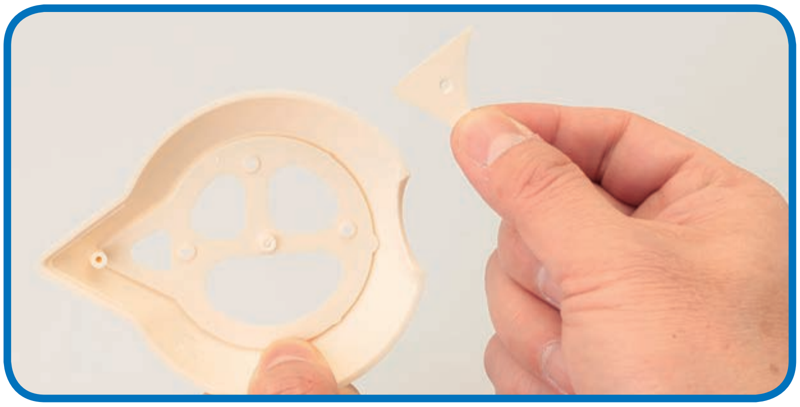
Right ear (Stage 4)

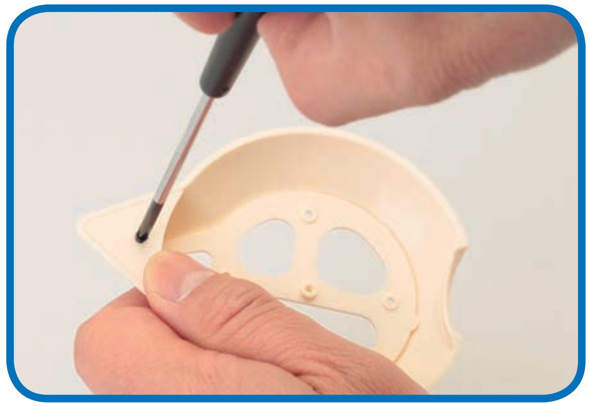
Head (Stage 4)



Neck stand (Stage 6)

ASSEMBLING THE LEFT EAR

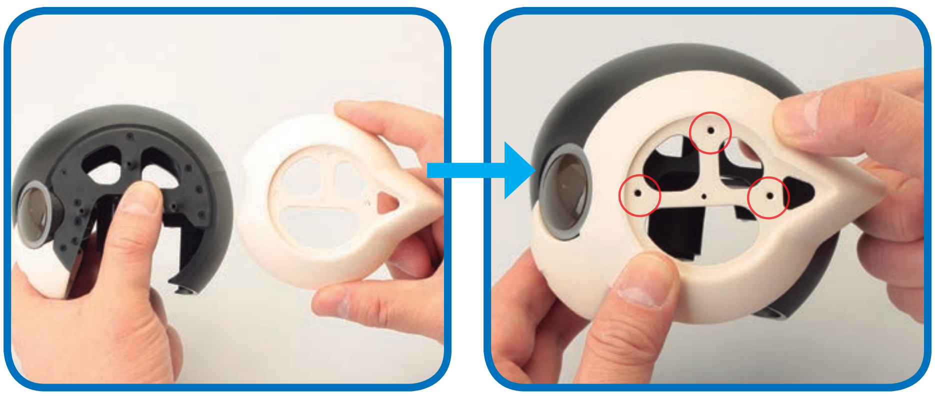




1 Line up the left ear backing with the inside of the left ear base. The countersunk hole in the centre of the ear backing should be facing you.

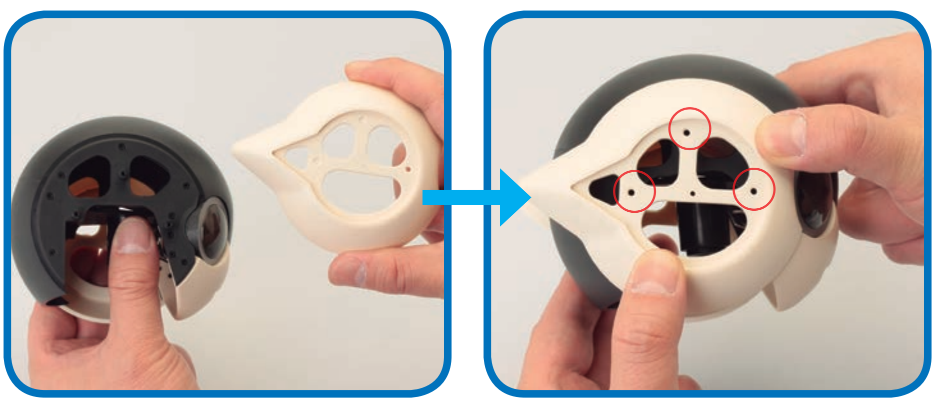
2 Fit an M2 x 4.5mm countersunk screw into the hole and tighten with the screwdriver to secure the parts.

ATTACHING THE EARS



3 Holding Robi's head frame so that he is facing left, line up the left ear assembly with the triangular point of Robi's ear facing directly behind him. Then, fit the parts together so that the three circled holes line up with those on the head frame.

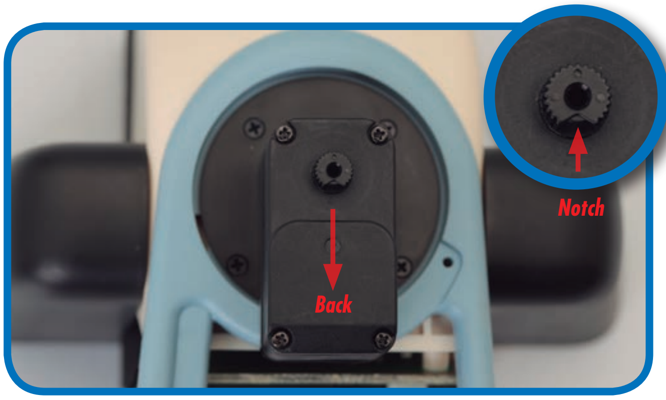
4 Hold the parts in place, and screw three of the M2 x 6mm screws into the holes to attach the ear to the head frame.

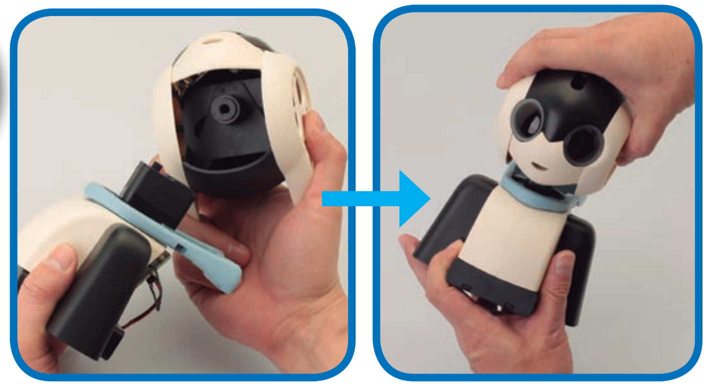


5 Turn Robi's head around, and line up the right ear assembly (from Stage 4) to the right head frame in the same way you did with the left in Step 3.

6 Fit another three M2 x 6mm screws into the corresponding holes to attach Robi's right ear.

ATTACHING THE HEAD





7 Take the neck stand you assembled in Stage 6 and locate the small notch in the servo shaft. This should be facing the back of the stand. Also note that the shaft has a series of splined ribs running up it.

8 With Robi's head facing the same way as the neck stand, fit it carefully into place so that the servo shaft slots into the small hole in the centre of the head frame. This has internal ribs that need to line up with the splines, so ensure the head is straight.

Assembled neck stand and servo cable

The initial phase of assembling Robi's head is now complete, and it is ready for testing – see below for details.

KEEP THE PARTS TIDY

Store the unused forearm safely until it is needed.

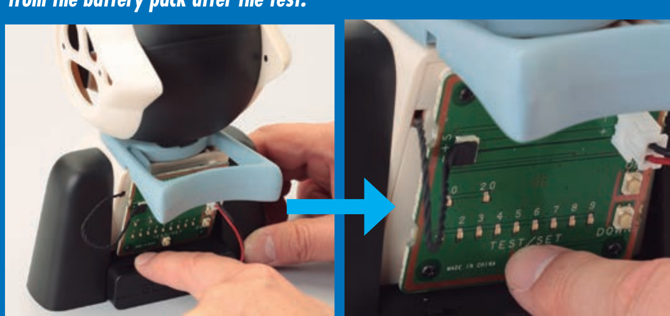
CAUTION!

Never pick up the assembly by the head, as this may strain or damage the servo or other parts.

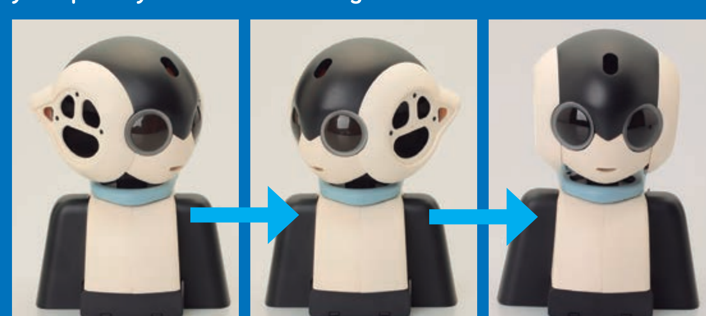
THE HEAD TEST

Robi's head movement is now ready to be tested. As you did when you tested the servo in the previous stage, first turn on the power from the battery pack. Then simply press the TEST/SET switch on the test board and watch as Robi moves his head from side to side. Robi's head may shake a little during this initial test, but don't worry; this is because it is only temporarily attached at this stage.

Don't forget to turn off the power supply from the battery pack after the test.



Turn the power on. *Press the TEST/SET switch.*



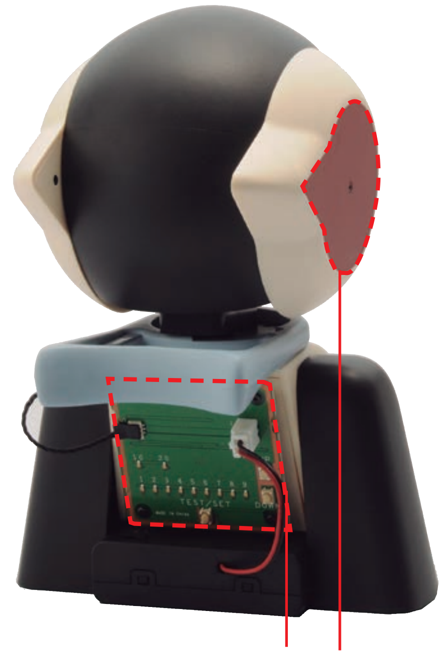
Left *Right* *Stop*

STAGE 8: FIT ROBI'S EAR TRIMS AND SET UP AN ARM SERVO

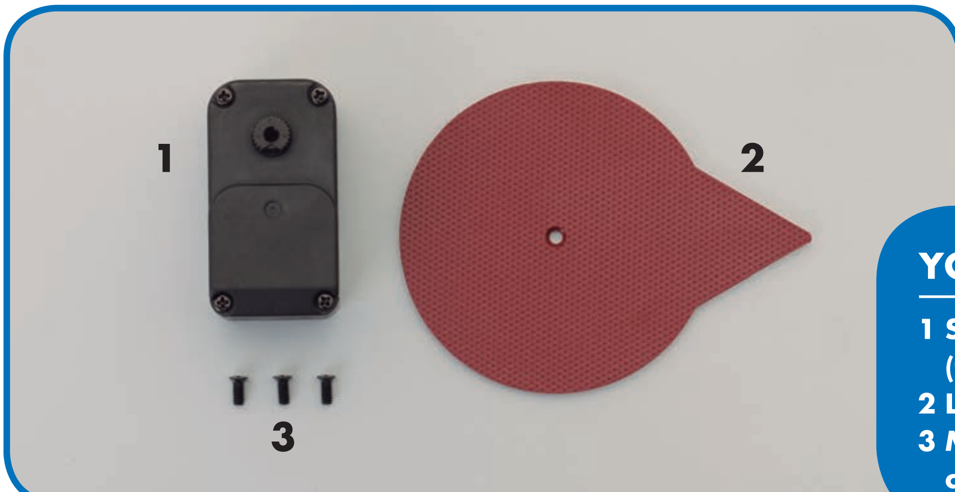
Add the two ear trim panels to complete the outer casing of Robi's head. Then prepare the servo motor that operates Robi's right forearm, and give it the correct identification (ID) number.

To move his arms, legs and head, Robi is fitted with 20 servo motors (a diagram showing these is provided in this stage). Robi's operating program needs to know which is which, so each servo has its own ID number – from 2 to 21. For example, the servo that moves his right forearm is number 18.

When first supplied, all the servos have the same ID number, ID=1, so you need to change each one according to where it will fit. To set the ID numbers, you use the test board. This may seem a little bit complicated the first time, but it is simple to master if you follow the instructions carefully.



PARTS TO BE ASSEMBLED



YOUR PARTS

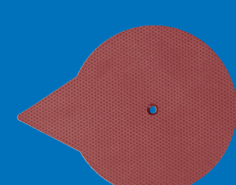
1 Servo motor (for right elbow)
2 Left ear trim panel
3 M2 x 4.5mm countersunk screws (x 3)

SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.

70mm servo cable (Stage 6)

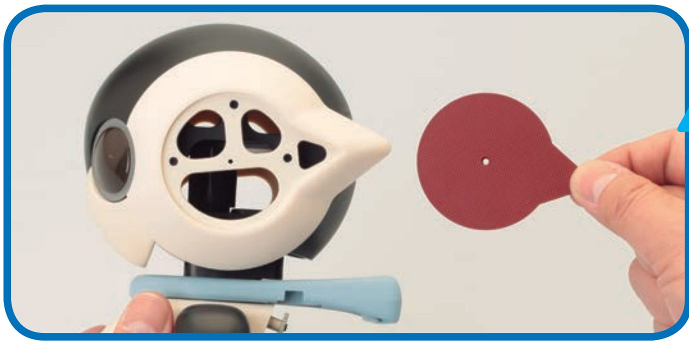
Head assembly (Stage 7)



Right ear trim panel (Stage 1)

Protective pads (Stage 3)

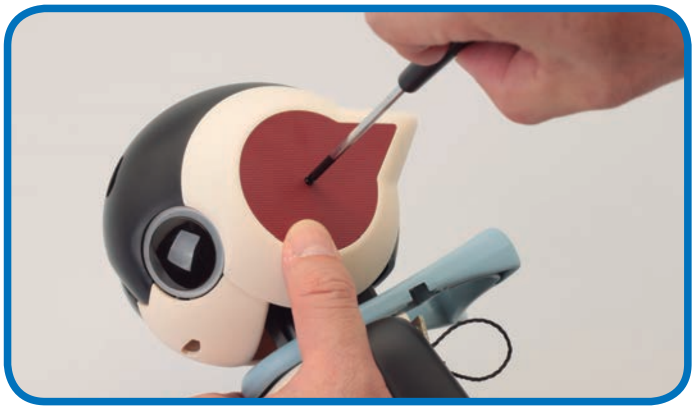
FITTING THE EAR TRIM PANELS



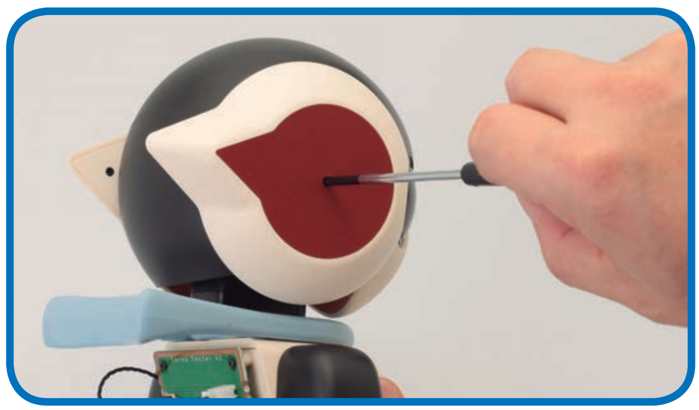
1 Take the head assembly from Stage 7 and the left ear trim provided with this stage.

TIP!

The outer surface has a rough texture. The smooth surface is designed to face inwards.

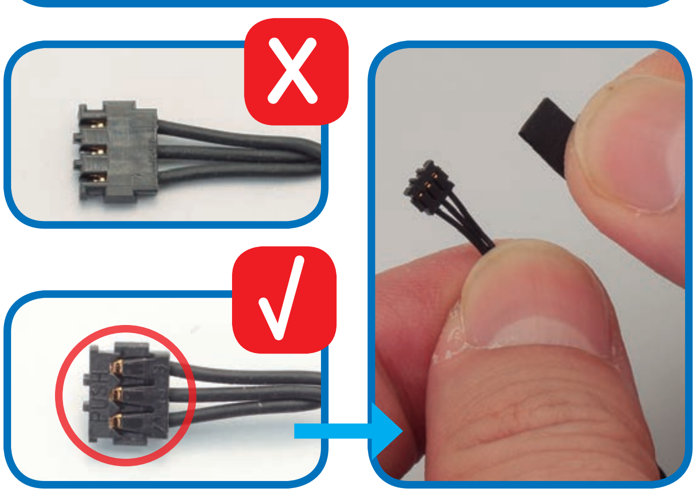


2 Align the trim with the recess and secure it with one M2 x 4.5mm screw. Do not overtighten the screw.



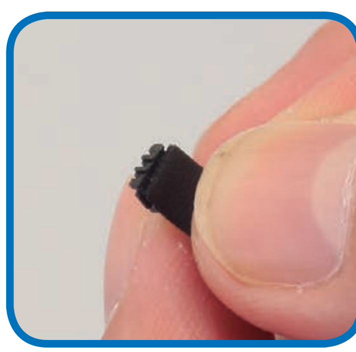
3 Repeat Steps 1 and 2 to fit the right ear trim panel that was provided with Stage 1.

SEALING THE SERVO CABLE



4 As you did in Stage 3, inspect the connectors at the ends of the servo cable. You need to apply a protective pad to the side with PUSH marked on it (circled).

TIP!



You may find it easier to apply the protective pads after fitting the cable to the servo at Step 8. If you do this, make sure you get the connector the right way up, with PUSH on the outside when you fit it to the servo.

5 The pad should be set centrally, and not extend beyond the connector. Press the pad firmly into place to seal the connector. Repeat at the other end of the cable.

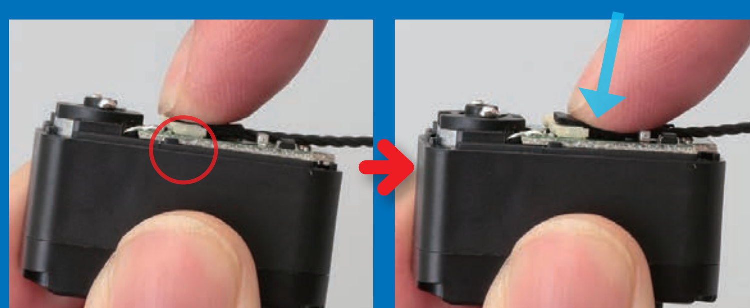
HANDLE WITH CARE!

The servo cables carry the electrical signals that operate Robi's motors. It is essential that each one is connected properly, and not placed under strain. If a cable is damaged, you may find that one of Robi's motors won't work at all, or stops working after a time.

Check the contacts

Make sure that each connector on the end of a cable is pushed all the way home into its socket. Otherwise, you may find that it works loose in use, due to the movement of the servo.

Even if the connector appears to have made contact, you may find that the base is not properly engaged with the socket, and can work loose.



Ensure that the connector is pushed all the way forwards and all the way down.

Do not pinch the cable

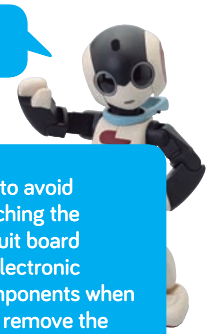
Ensure that the cable is not trapped between the parts, or hooked around screws and other cables. Otherwise, there is a risk that it may be pulled loose, chafed, or even broken, causing an electrical short circuit. Whenever you install a servo, check the assembly instructions carefully to see how to route the cables.

Use the protective pads

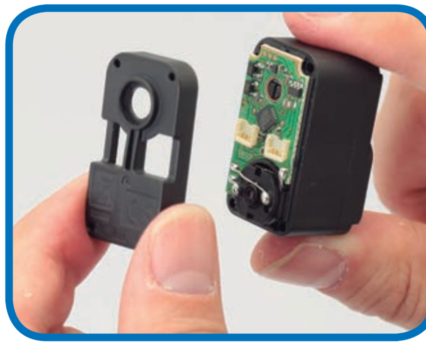
The protective pads play an important role in stopping the connector and cable from being strained and possibly broken. Always fit them following the assembly instructions carefully.

CONNECTING THE SERVO CABLE

BE CAREFUL!

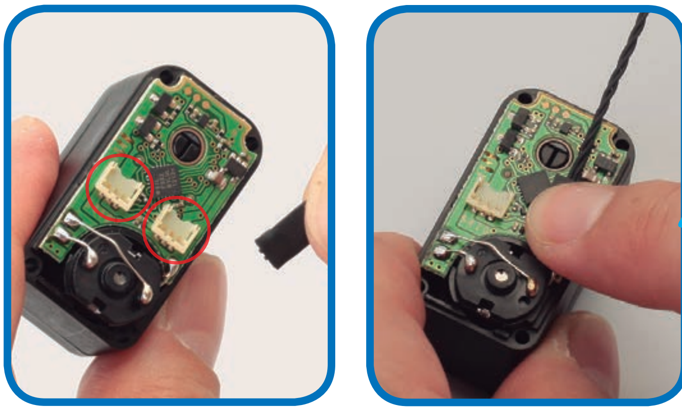


Try to avoid touching the circuit board or electronic components when you remove the cover from the servo.



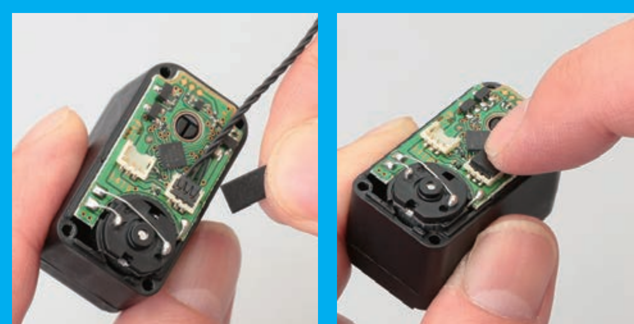
6 Use a screwdriver to undo the four screws securing the cover of the servo, and take each one out.

7 Carefully lift the cover away from the other side of the servo.

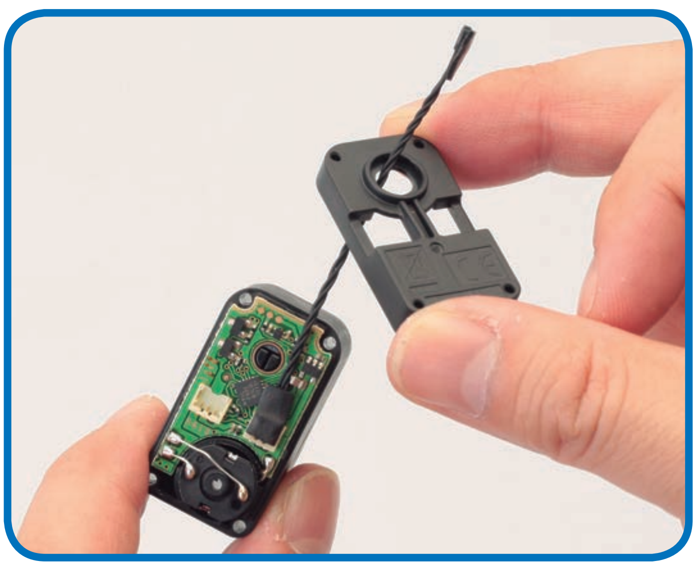


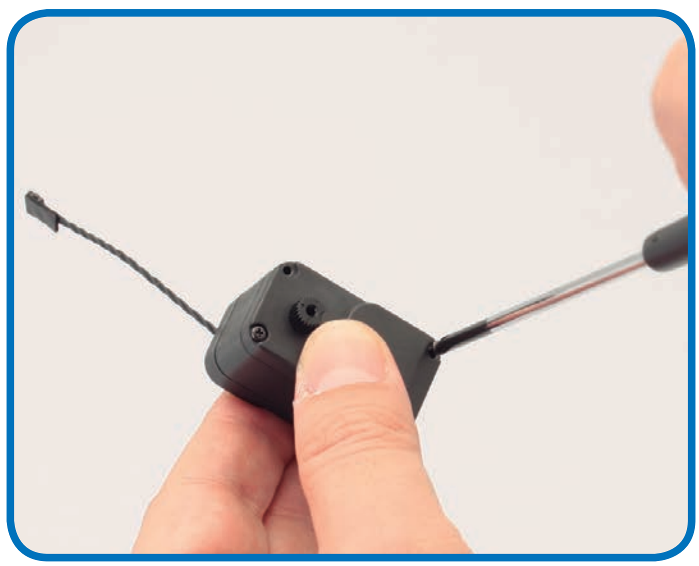
8 Take either end of the connector cable and carefully fit it in one of the two sockets circled in red. It does not matter which one. Gently push until it locks in place.

TIP!



If you chose not to fit the protective pads earlier (at Steps 4 and 5), you can fit it now, which will help to secure the connector in place.





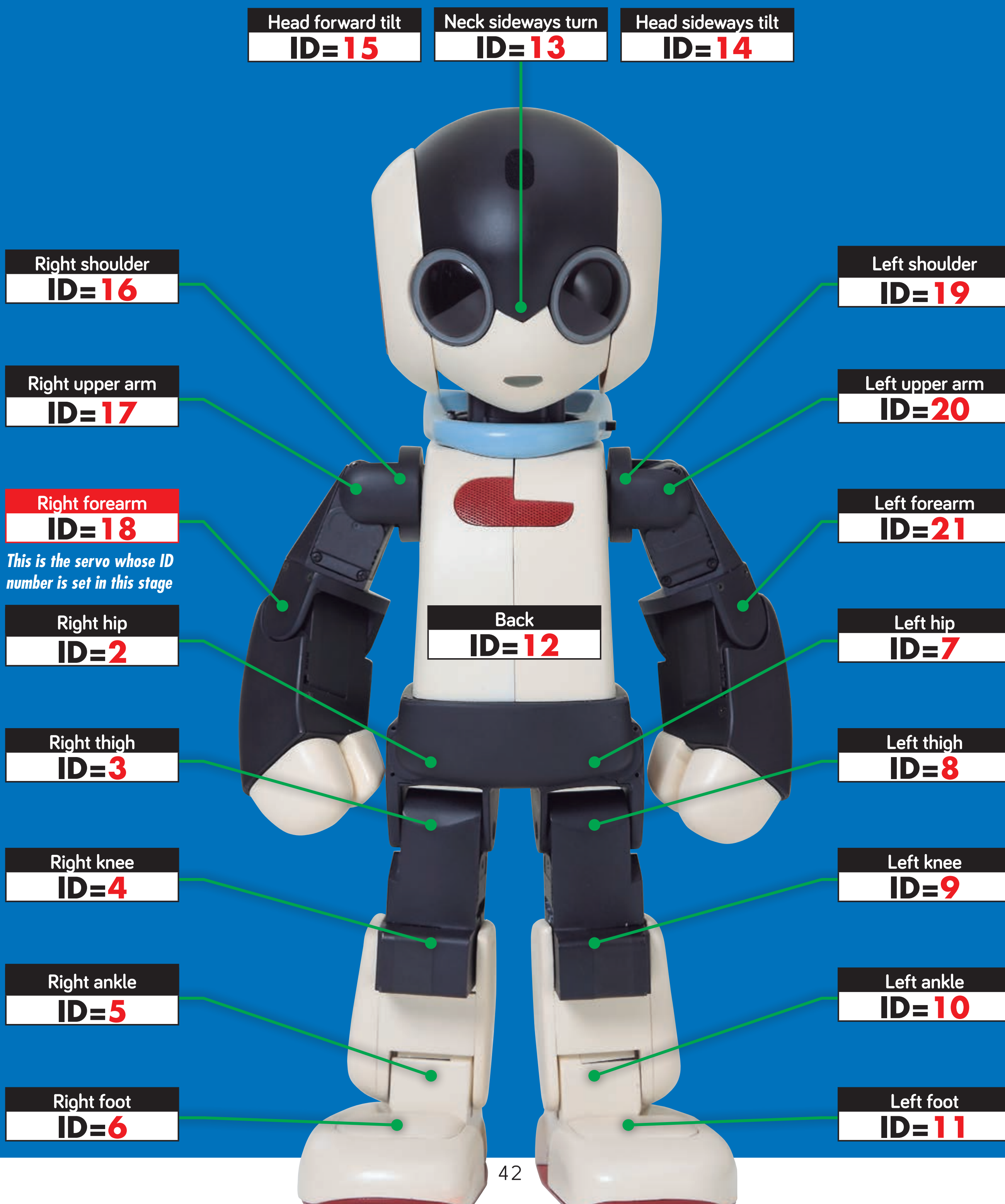
9 Carefully feed the free end of the cable up through the circular hole in the servo cover. Make sure that the cover is the right way up.

10 Fit the cover onto the servo casing, then turn the servo over and insert the four screws you removed in Step 6. Tighten them carefully to secure the cover.

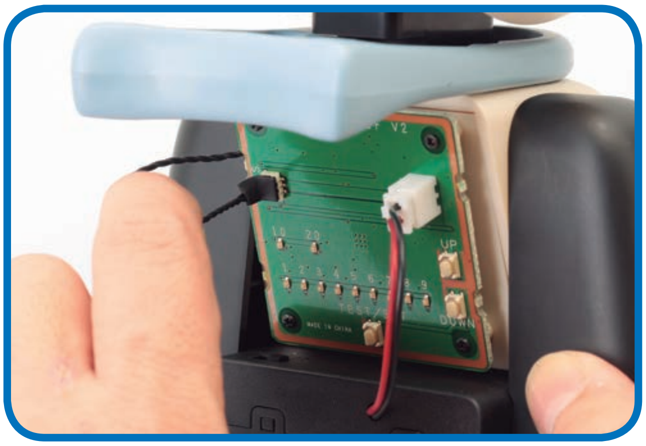
SERVO ID NUMBERS

CLOSE-UP

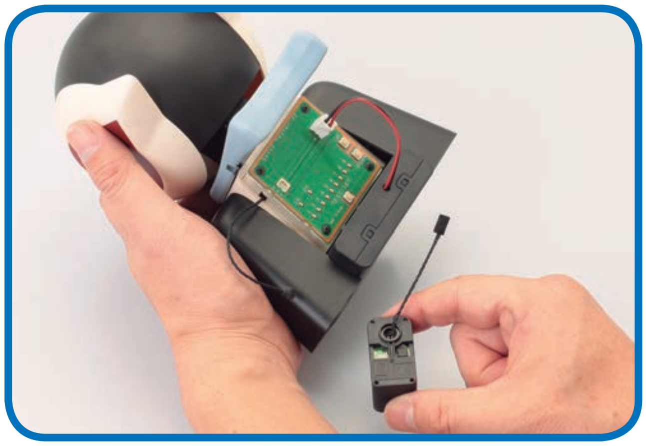
Each of Robi's servo motors is given a unique ID number. If one has the wrong ID, Robi will not work properly – but there is no need to worry, because for every servo you receive, the Assembly Guides will show you how to set and check its number. Here's a guide to all Robi's servos and the numbers they will be given. This is done by using the test board mounted in the neck stand.



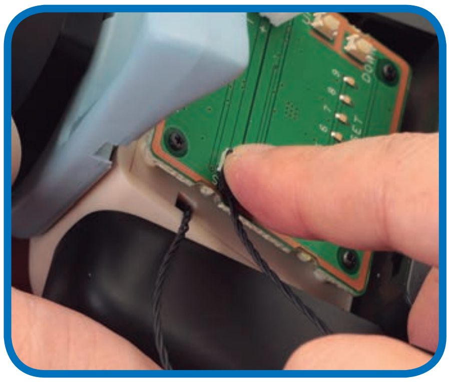
CONNECTING THE SERVO



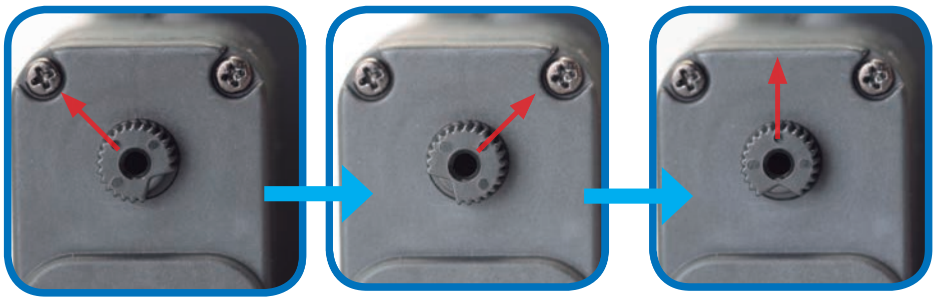
11 Take the neck stand and make sure the power pack is turned OFF. Carefully unplug the black servo cable from the socket on the left of the test board.



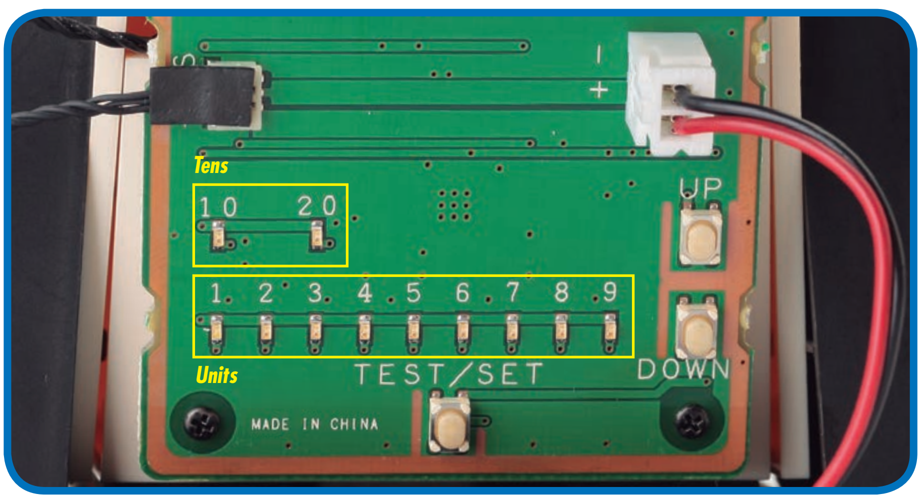
12 Take the right forearm servo with its cable attached.



13 Plug the end of the right forearm servo cable into the socket you just removed the cable from.



14 Press the TEST/SET switch briefly to run the servo test. The shaft at the top of the servo casing should rotate 45 degrees to the left, stop, then move back across to 45 degrees to the right. After you have run the test, turn off the power switch on the battery pack.

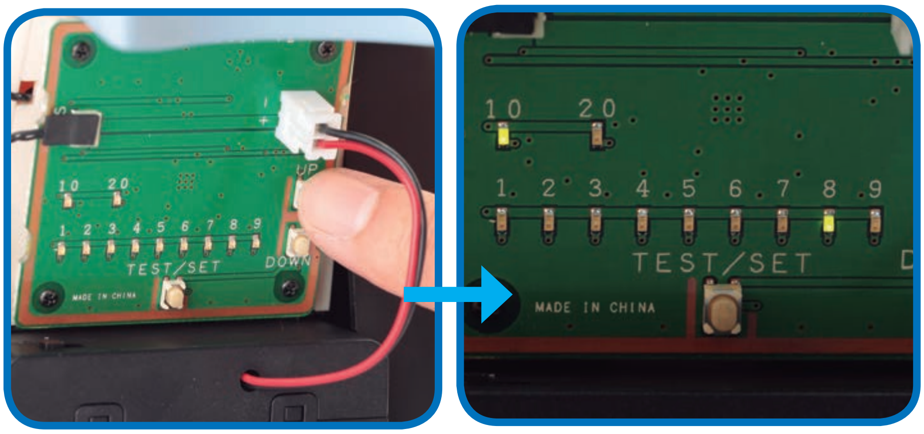


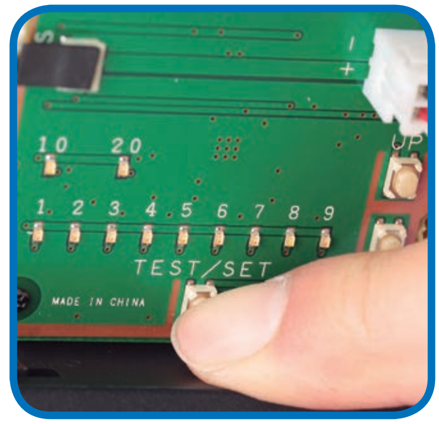
15 Familiarise yourself with the lights and switches on the test board. These are used to set the ID number. Each time you press the UP switch, the ID number increases by one, and the result is displayed by the two banks of LED lights. The top row reads in tens and the bottom in units. For example, to set the ID to 18, press UP 17 times. This will light up the 10 LED and the 8 LED. (If you overshoot, press the DOWN switch to take it down one step at a time.) When the ID number is correct, you set it by pressing and holding the TEST/SET switch.

HELP!

If the LEDs flash rapidly when you run the test at Step 14, you have a faulty connection. Double-check the servo cable's connection to the socket on the test board, TURNING THE POWER PACK OFF BEFORE YOU DO. If the light still flashes, carefully unscrew the servo housing and check that the internal connector plug is in place. If you pressed the wrong button, reset the board by holding down the TEST/SET switch for several seconds.

SETTING THE ID NUMBER

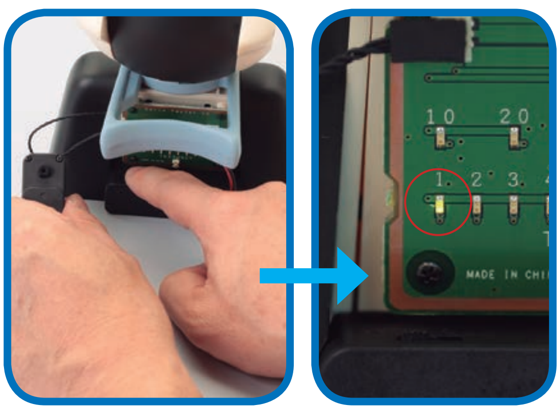




16 To set the ID number of the right forearm servo to 18, press the UP button 17 times. Confirm the setting by checking that the 10 and 8 LEDs are lit, using the UP and DOWN buttons to correct the number if necessary.

17 Confirm the setting by pressing the TEST/SET button. The 10 and 8 lights should flash, then go out.

CHECKING THE ID NUMBER

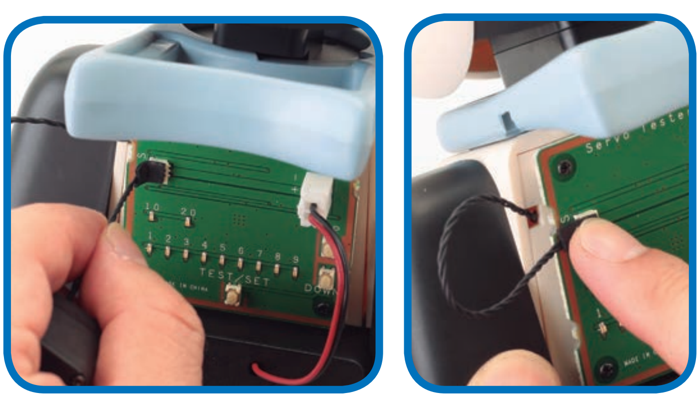


18 Leaving the servo connected, switch the power OFF, then back ON. Only the 1 LED should light up.

19 To check the ID number, press TEST/SET. Now the 10 and 8 should light up.

HELP!

It is normal for the 1 LED to light up when you switch on. It will only change to give you the ID number of the servo when you press TEST/SET. If you made a mistake and set the wrong ID number, you can reset it by going through the setting procedure again.



20 Once all is well, switch the power pack OFF, then carefully unplug the servo from the test board. It will retain its setting, ready to fit to Robi's arm. You can reconnect the servo lead that you removed in Step 11, but this is not essential.

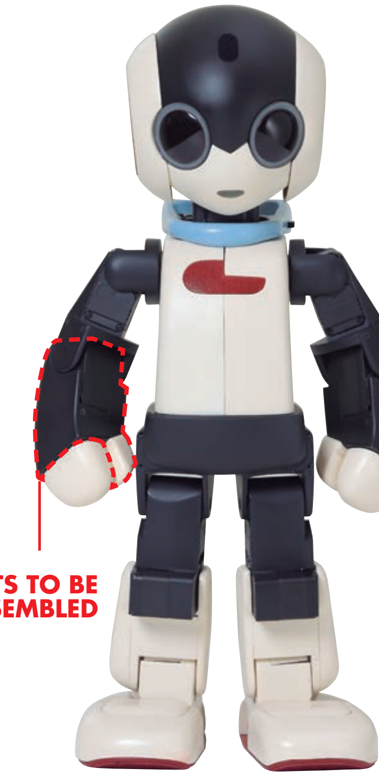
Assembled neck stand and forearm servo

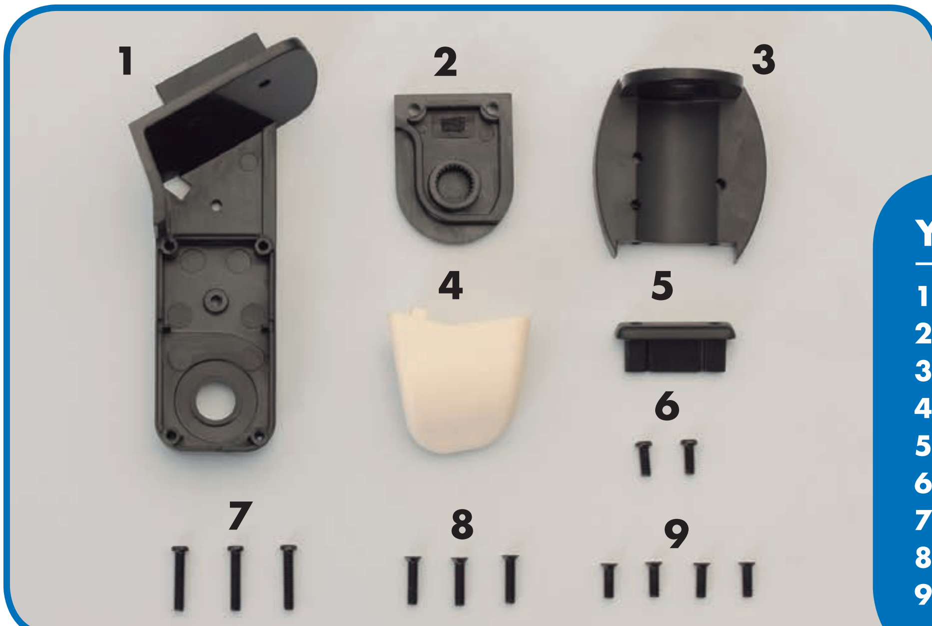
Robi's head casing is complete, and your first arm servo is ready for installation.

STAGE 9: BUILDING ROBI'S RIGHT FOREARM

In this stage you will assemble the main part of Robi's right forearm, and add his elbow and thumb. The assembly includes installing the servo motor that forms Robi's elbow joint.

To give Robi arm movements similar to a human, his elbow and shoulder joints are fitted with servo motors, and his right forearm is assembled around the elbow servo. But first, you will begin putting together Robi's hand. (Although Robi does not have movable wrist joints, his fingers and separate thumb are held together by a spring, allowing him to hold thin objects.)





YOUR PARTS

1 Right forearm frame
2 Servo front panel
3 Elbow joint
4 Right thumb
5 Right finger mount
6 M2 x 5mm pan-head screws (x 2)
7 M2 x 10mm pan-head screws (x 3)
8 M2 x 8mm countersunk screws (x 3)
9 M2 x 6mm countersunk screws (x 4)

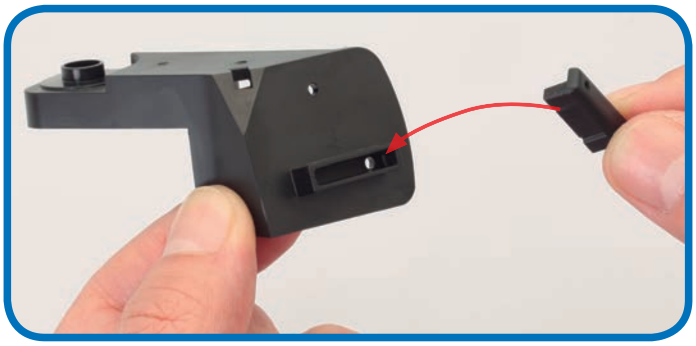
SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.

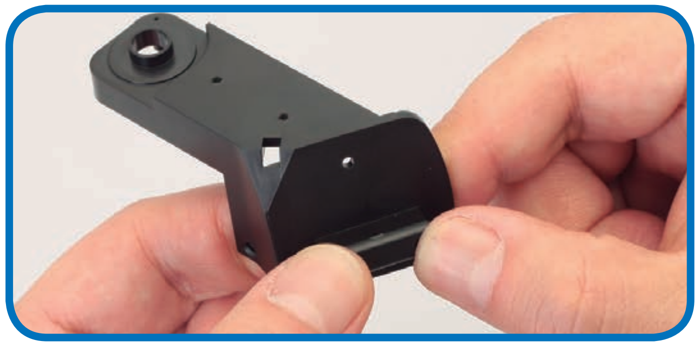
Right forearm (Stage 7)

Servo motor (assembled in Stage 8)

THE FINGER MOUNT

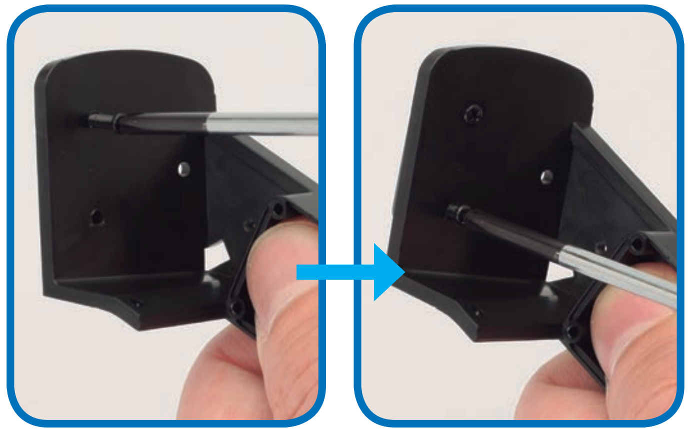


1 Align the right forearm frame and finger mount as shown. The mount can be fitted either way up.



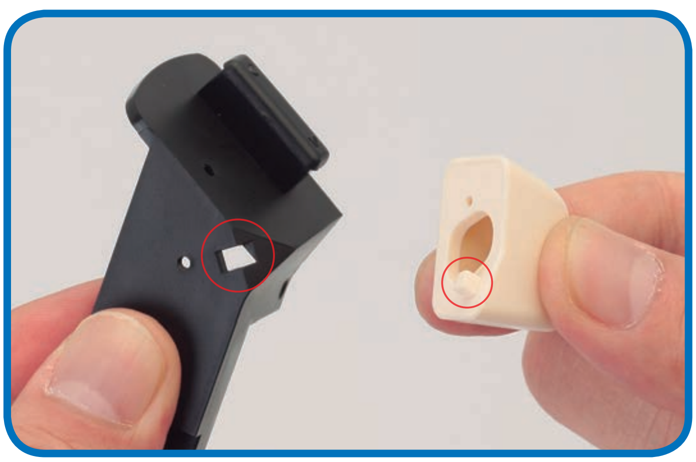
2 Press the tongue of the finger mount into the long slot in the end of the forearm frame.

3 Turn the forearm frame over to check that the holes in the finger mount are in line with the two screw holes in the forearm frame.

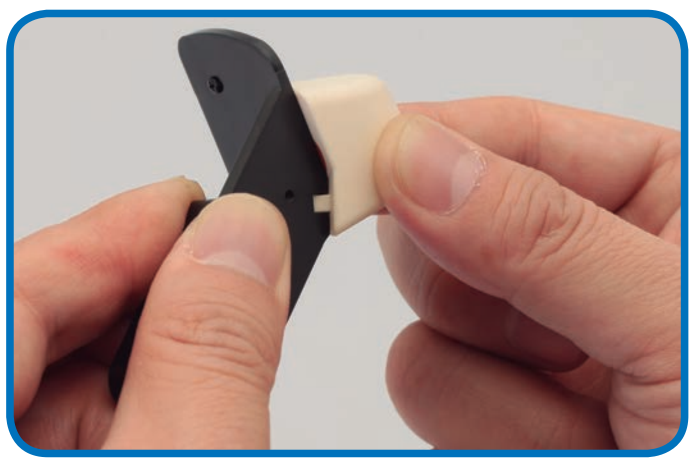


4 Fit two M2 x 10mm pan-head screws through the holes and carefully tighten them to secure the finger mount.

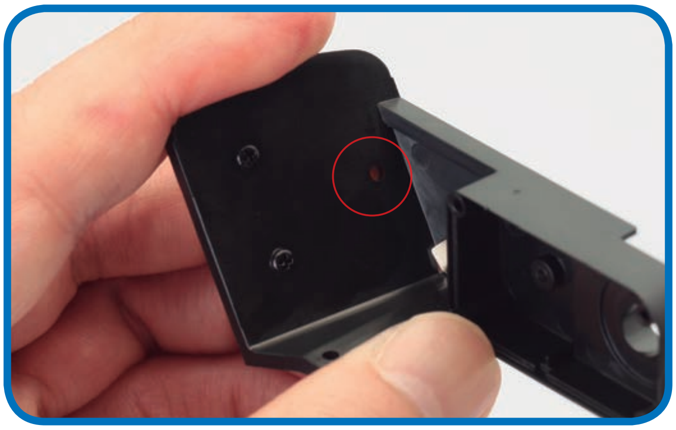
THE THUMB



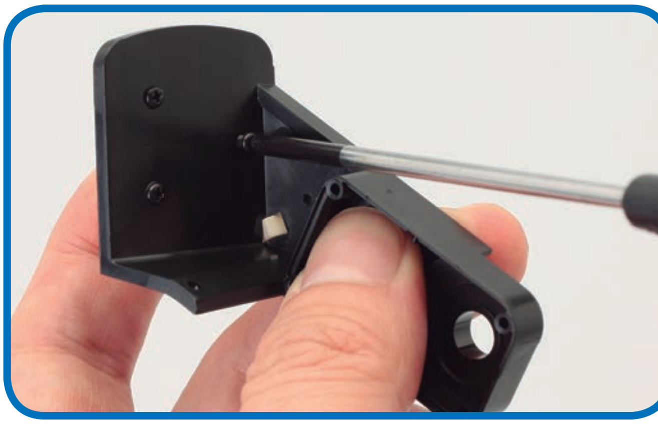
5 Align the thumb with the forearm frame, noting the position of the square tab on the thumb and matching hole in the frame, circled in red.



6 Press the parts together so that the tab locks into the hole and the thumb fits flush against the frame.

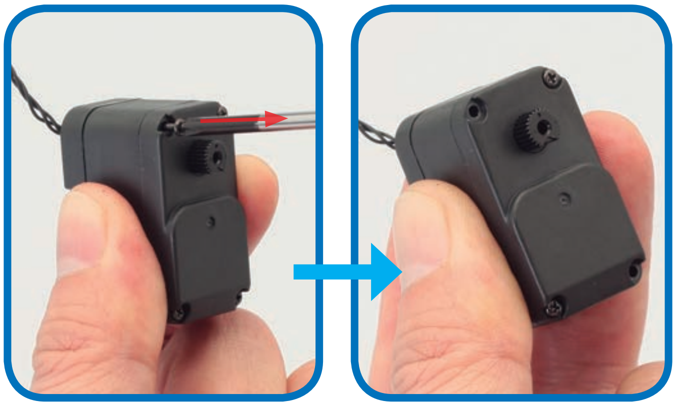


7 Turn the forearm frame over and check that the hole in the thumb is in line with the remaining screw hole in the forearm frame.

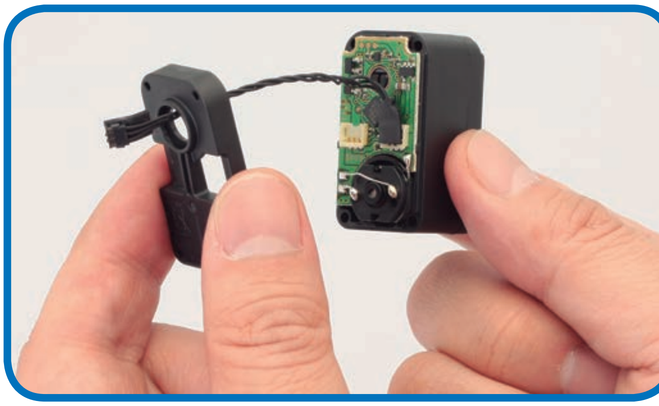


8 Fit an M2 x 5mm pan-head screw through the hole and carefully tighten it to secure the thumb.

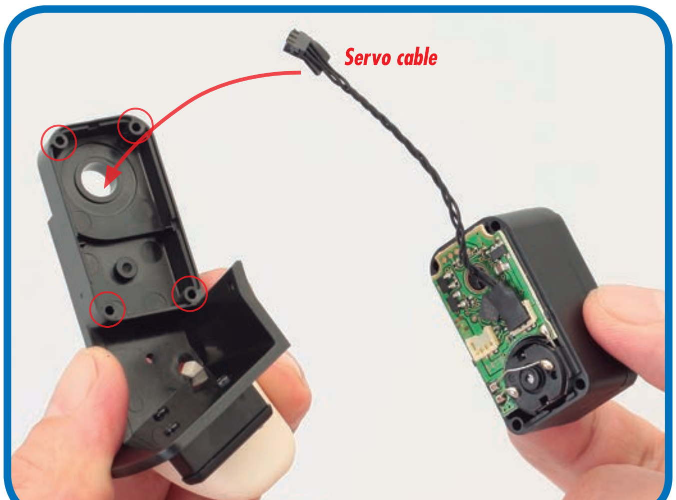
THE SERVO



9 Take the servo motor that you assembled in Stage 8, and remove all four long screws that secure the rear cover. Keep the screws safe as you will reuse them.

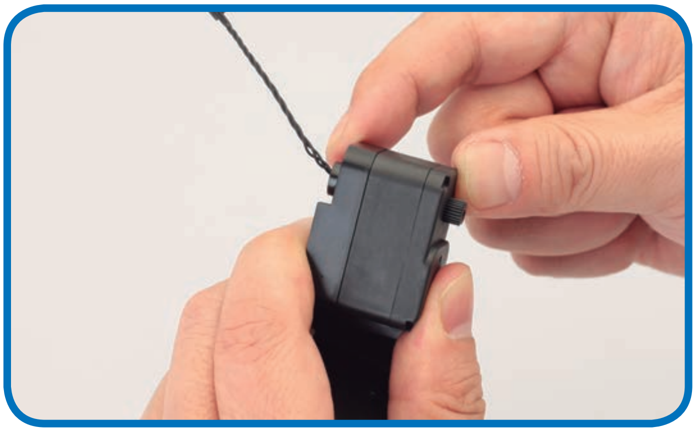


10 Carefully remove the servo cover. You will not be refitting this part, as the forearm frame fits in its place.

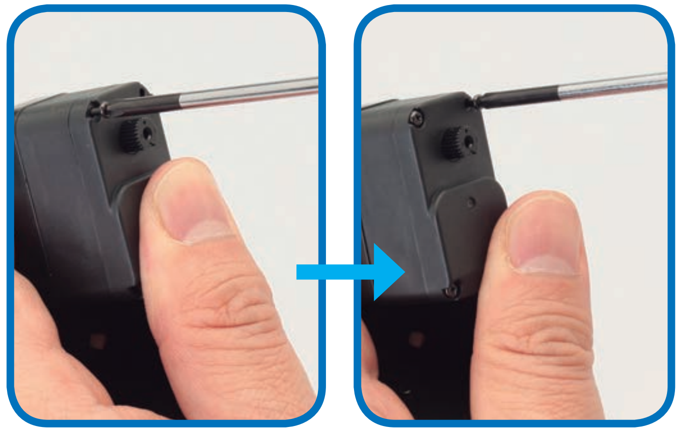


11 Align the servo motor with the right forearm frame as shown. You will need to feed the servo cable through the large circular hole in the forearm frame and align the screw holes in the servo with the four circled holes.

12 Carefully feed the servo cable through the hole in the forearm frame.

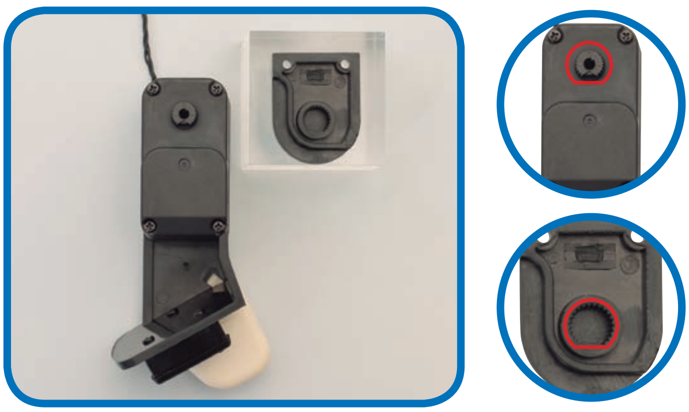


13 Press the servo into place, making sure it fits flush and that you do not trap the lead.



14 Using the four screws you removed in Step 9, secure the servo to the forearm frame.

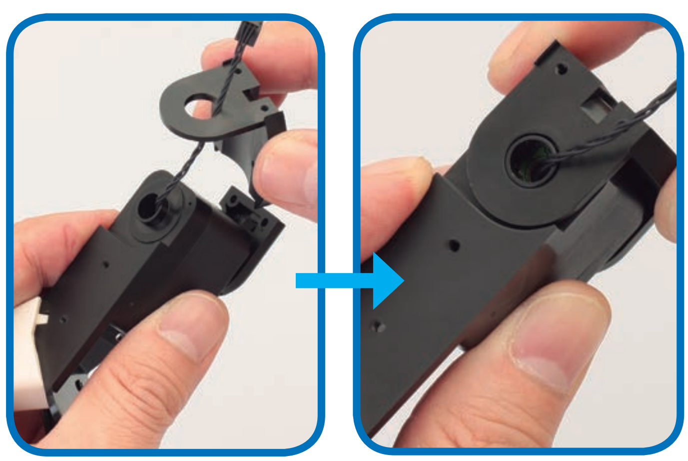
ELBOW JOINT



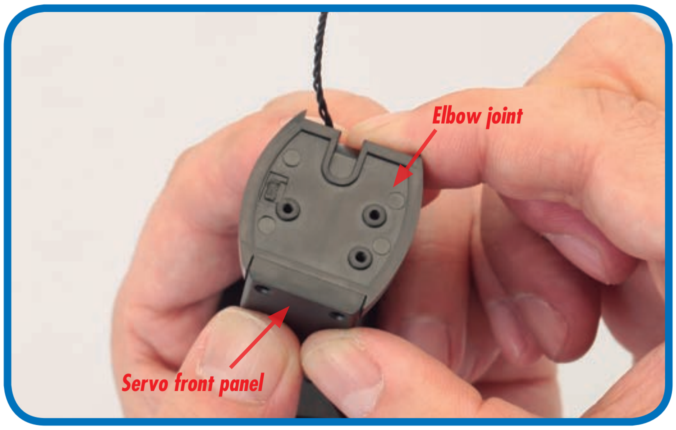
15 If you examine the servo front panel you will see that it has a recess that matches the shaft projecting from the servo. Both have a flat section, making them slightly 'D'-shaped, as outlined in red.

16 Align the flat on the recess in the servo front panel with the flat on the servo shaft.

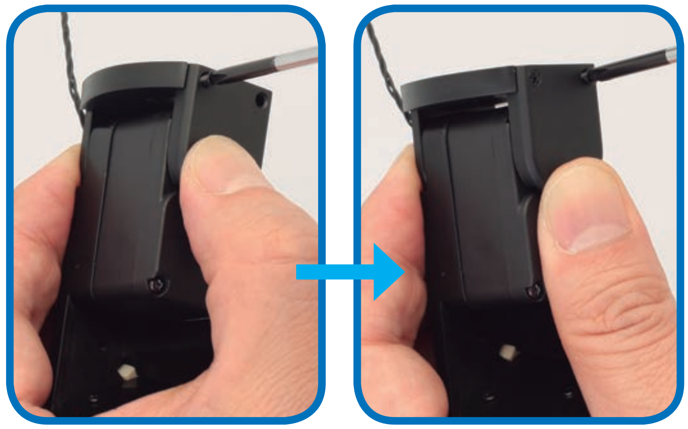
17 Press the front panel onto the shaft. The parts are quite a tight fit, but if the front panel won't clip into place, check that it is aligned correctly.



18 Fit the elbow joint over the servo as shown. Carefully feed the servo cable through the large hole in the elbow joint, and then fit the hole over the circular collar projecting from the right forearm frame.

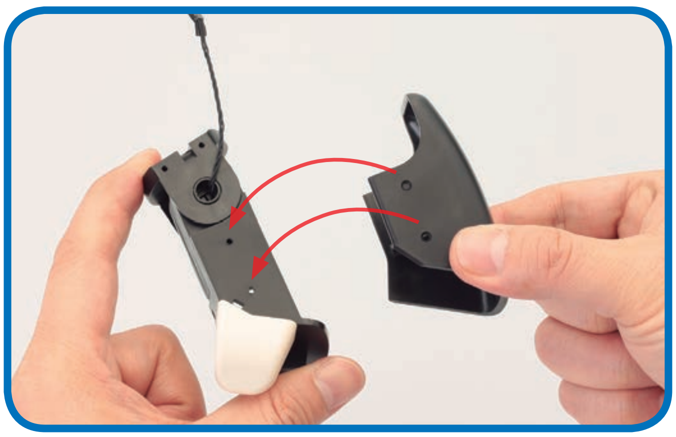


19 Ensure that the servo front panel fits into the recess in the elbow joint, as shown.



20 Use two M2 x 8mm countersunk screws to fix the front panel to the elbow joint.

THE RIGHT FOREARM



21 Take the right forearm and fit it over the assembly as shown, so that the screw holes match up.

22 With the forearm fitted in place, the assembly should look like this.

23 Fit two M2 x 6mm countersunk screws into the holes shown in the previous step, then turn the forearm over and fit another screw into the remaining hole under the forearm frame to complete the assembly.

Assembled forearm

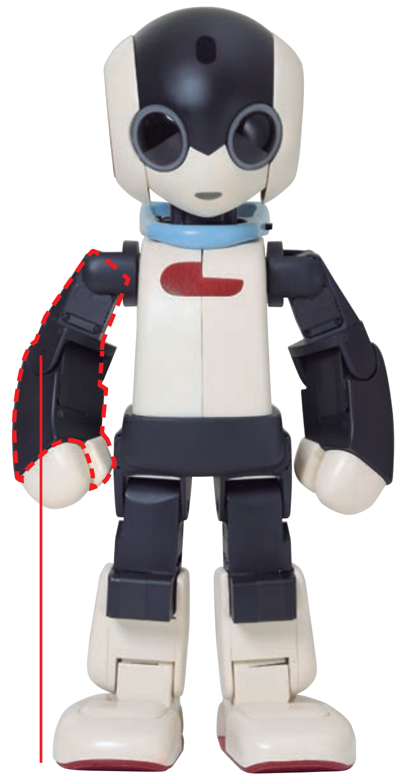
This stage is now complete, and Robi's right forearm is beginning to take shape.

ASSEMBLY GUIDE

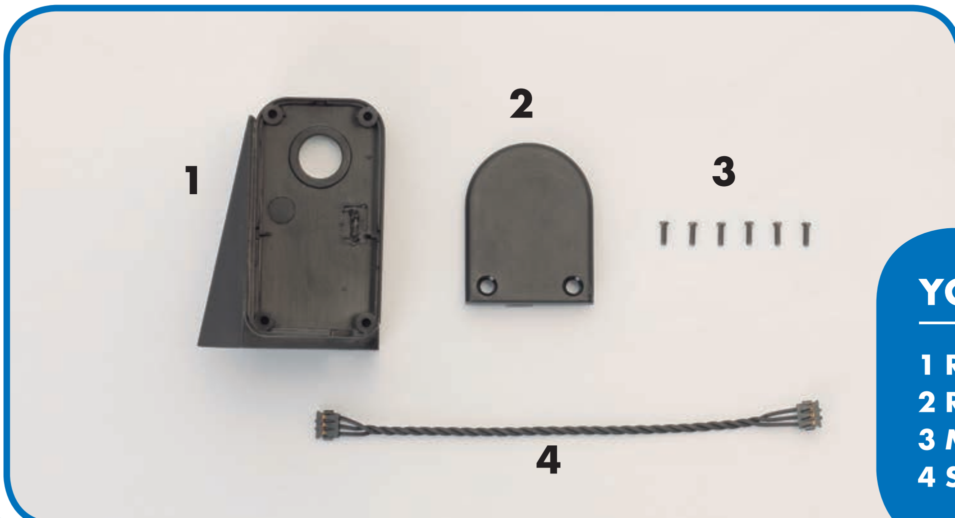
STAGE 10: CONTINUE BUILDING ROBI'S RIGHT ARM

In this session, you will attach Robi's right upper arm frame, which will hold the servo used to give him shoulder movement, to the right forearm assembled in the previous stage.

The upper arm frame will join to the forearm assembly at the elbow, which you will seal off from the rear with the elbow back panel also supplied with this stage. As you do this, you will feed the forearm's servo cable through a small trough set into the side of the back panel and frame. It is very important that the cable fits through this recess freely, without getting trapped or caught between either part, as this may damage the cable and later affect the operation of Robi's arm. Lastly, you will prepare this stage's servo cable for safe use, as you have done in previous stages.



PARTS TO BE ASSEMBLED



YOUR PARTS

1 Right upper arm frame
2 Right elbow back panel
3 M2 x 6mm flat-head screws (x 6)
4 Servo cable (70mm long)

SAVED PARTS

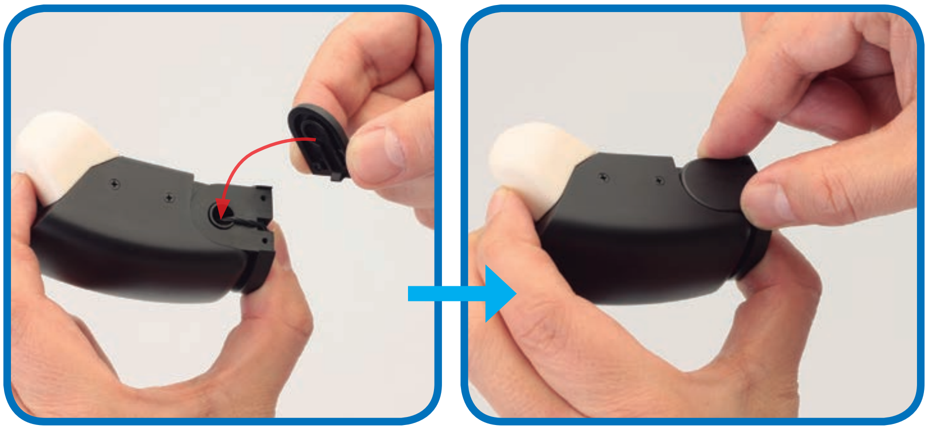
You will be using parts from the previous stages, so be sure to have these to hand before beginning.

Right forearm assembly (Stage 9)

Protective pads (Stage 3)

FITTING THE RIGHT ELBOW BACK PANEL

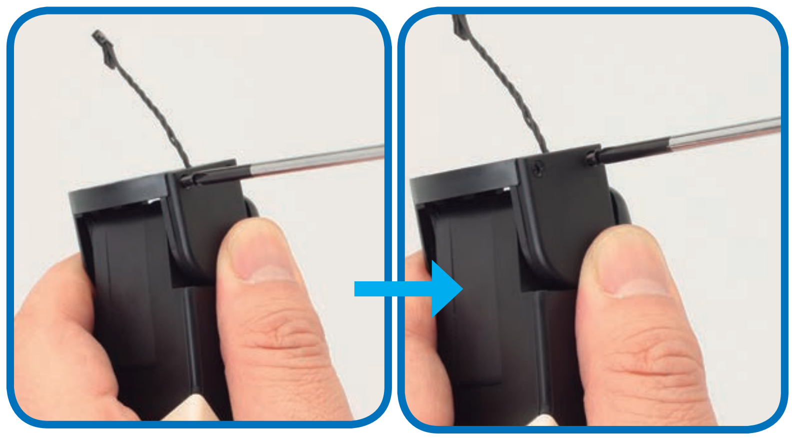
1 Take the right forearm assembly (Stage 9) in your hands and gently pull the protruding servo cable along so that it rests in the recess at the elbow, as shown.



2 Holding the servo cable in place within the recess, press the right elbow back panel into position, so that its curved edge matches that of the forearm (see red arrow).

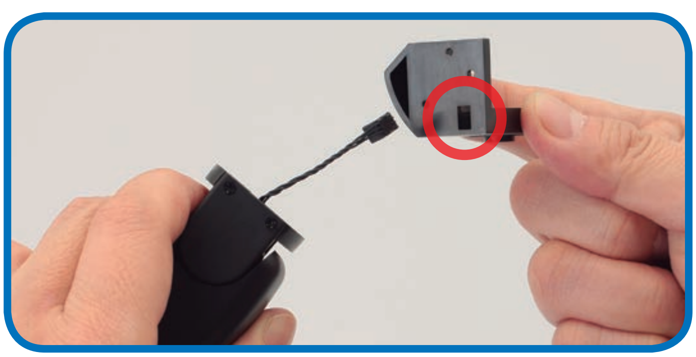
3 Press the parts together firmly between your fingers to make sure they are securely joined.

Make absolutely sure the servo cable is sitting in the recess and not trapped between the parts before you proceed!



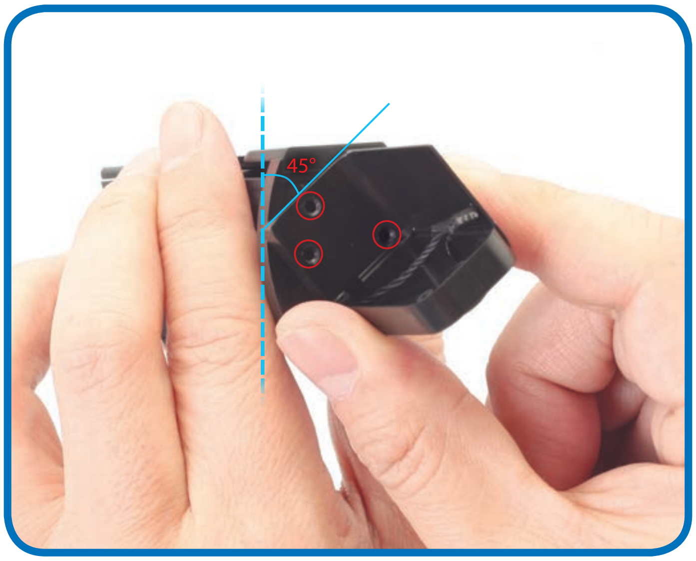
4 With two of the M2 x 6mm flat-head screws, fasten the parts together with the screwdriver supplied with Stage 2.

ATTACHING THE RIGHT UPPER ARM



5 Line up the forearm assembly to the right upper arm frame, noting the position of the rectangular hole (circled).

6 Carefully pass the tip of the servo cable through the rectangular hole, as shown.



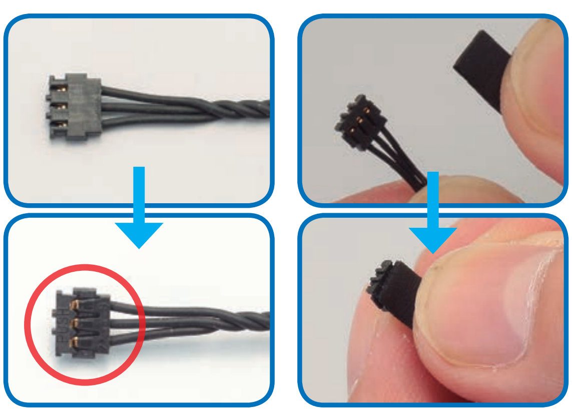
7 With the parts held at a 45° angle to one another, line up the three circled screw holes with those of the forearm beneath.

8 Once you have the screw holes lined up, hold the parts together in one hand, making sure that they do not slip out of position.

Again, make sure the servo cable is still sitting in the recess and that it is not trapped between the parts in any way before you tighten the three screws!

9 Still holding the parts in one hand, tighten three more of the M2 x 6mm flat-head screws into the holes lined up in the previous step.

PREPARING THE SERVO CABLE



10 As you have done in previous stages, identify the side of the servo cable's connector marked 'PUSH'. Then, separate a protective pad from the sheet (Stage 3) and place this on top of the connector, making sure that the pad does not extend beyond the connector's tip. Repeat for the connector at the other end of the servo cable so that both are covered.

Assembled right arm and servo cable

Robi's right arm is nearly complete, ready to be fitted with another servo to operate his shoulder movement, and your next servo cable is protected and ready for use in the coming stages.

COMING IN PACK 4

Your next FOUR complete Stages, as Robi starts to come to life!

Stage 11
Add another servo to Robi's right arm

PARTS PROVIDED
The servo that will operate Robi's right upper arm.

Stage 12
Adding Robi's right shoulder joint

PARTS PROVIDED
Components to build up Robi's shoulder joint, along with another servo cable.

Stage 13
Preparing Robi's shoulder servo

PARTS PROVIDED
The servo that will operate Robi's right shoulder.

Stage 14
Begin assembling Robi's body

PARTS PROVIDED
Components to build up Robi's body, including the right body cover and servo horn.